

HISPANO-SUIZA MODEL 8BE 220KP V8

Dates produced	1916 onward
Displacement	718 cu in (11.8 liters)
Max power output	220 hp at 1,600 rpm

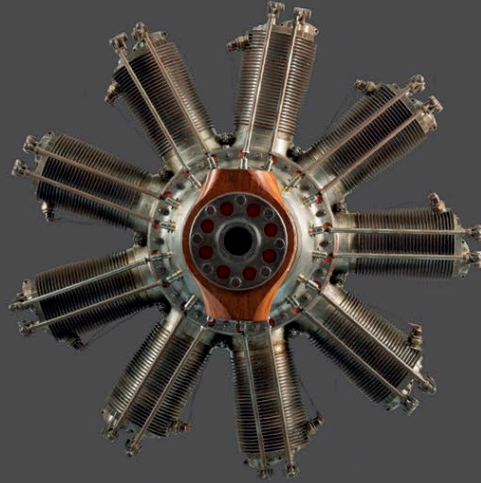
Swiss engineer Mark Birgikt's Hispano-Suiza V8 has been judged the design that "showed the world how a water-cooled engine ought to be built." Its pistons ran in steel liners screwed into cast aluminum blocks, the valve gear being fully enclosed and protected from dust. Rigid, light, and durable—at least when built by Hispano-Suiza, for its construction challenged precision standards of the day—the "Hisso" influenced all the big "V" engines that followed.



CLERGET 9B

Dates produced	1917-1918
Displacement	992 cu in (16.3 liters)
Max power output	130 hp at 1,250 rpm

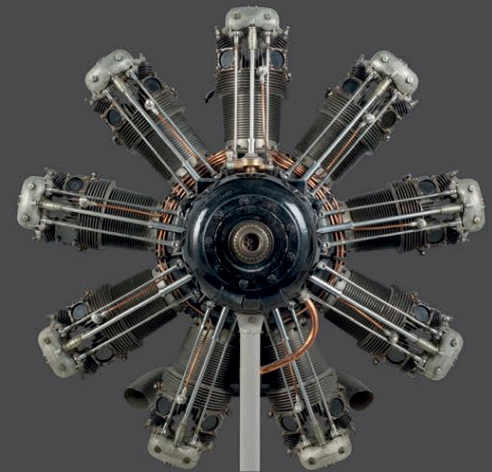
The Clerget was a rotary—an engine that spun with the propeller to aid cooling in the days of low-quality fuel and hot-running cylinders. The standard engine of the Sopwith Camel, it was distinguished by conventional pushrod and rocker valve gear and was the inspiration for W.O. Bentley's much more powerful BR1 and BR2, regarded by many as the ultimate rotary engines. After WWI, rotaries soon gave way to the more reliable and economical radial engine.



BRISTOL JUPITER

Dates produced	1920-1935
Displacement	1,752 cu in (28.7 liters)
Max power output	580 hp at 2,200 rpm

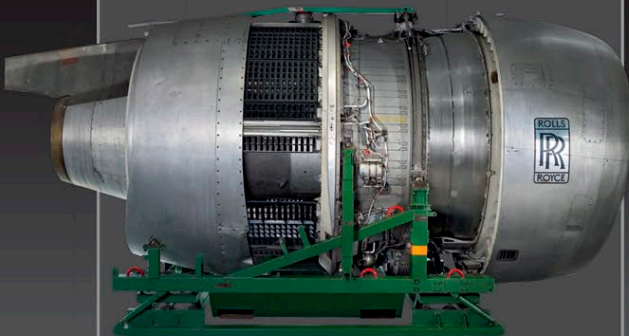
The Jupiter first appeared as a Cosmos Engineering product that missed out on service during the final year of World War I thanks to the government backing an inferior rival. Under Bristol ownership, it was developed by Roy Fedden to become one of the finest and most extensively foreign-licensed radial engines of the 1920s and '30s. The use of four-valve heads—which gave good "breathing"—and close attention to design detail ensured high power with great reliability.



ROLLS-ROYCE RB211

Dates produced	1972-present
Engine weight	9,670 lb (4,386 kg)
Max thrust	60,600 lb (270 kN)

Originally developed for the Lockheed L-1011 Tristar, the RB211 was famously the engine that both launched Rolls-Royce as a leading manufacturer of commercial high-bypass turbofans and brought the company to bankruptcy, before it was refloated by the British government. It was the first triple-spool turbofan, having three groups of turbines, each group driving an individual compressor via a concentric shaft. This complex construction gave improved fuel efficiency.



BOEING GE90-115B

Dates produced	1995-present
Engine weight	18,260 lb (8,283 kg)
Max thrust	115,300 lb (513 kN)

Used in the Boeing 777, which on two engines performs the kind of long-range over water passenger operations that used to be confined to four-engine jets, the GE90 is one of the largest, most powerful, and reliable turbofans ever built. Developed from the 1970s NASA Energy Efficient Engine, it has set records for the biggest fan diameter (10 ft 7 in / 3.25 m); highest compressor pressure ratio (23:1, against the 4:1 of early jet engines); and greatest thrust (127,900 lb/569 kN).



CENTURION 2.0

Dates produced	2006-present
Displacement	122 cu in (2.0 liters)
Max power output	155 hp at 4,200 rpm

The Centurion diesel range owes its existence to German motor racing supplier Frank Thielert seeing the potential for adapting the advanced automotive diesel engines manufactured by Mercedes-Benz for use in light aircraft. The initial version was a 1.7-liter unit that produced 135hp; it was noted for its fuel economy but lacked power in comparison with traditional avgas engines. The unit now manufactured by Centurion has the same overall dimensions but produces more power.

